Separating forcing and low-frequency feeding in a signed Navier–Stokes tail budget

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Abstract

For a smooth finite-energy forced Navier–Stokes evolution, we bound the absolute time integrals of direct force work and low-frequency feeding in the high-frequency enstrophy balance. The bounds use kinetic energy, total energy dissipation, a fixed cutoff, and derivatives of the force; they spend none of the high-frequency viscous dissipation. Consequently, for the forced singularities constructed in [1], the accumulated tail self-stretching minus even the full tail dissipation is unbounded at every fixed cutoff. The full-field obstruction is checked in Lean. The Fourier-tail extension below is an analytic proof, not a formalized projector theorem. We also bound the forcing derivative of a quartic caloric stretching functional using only energy-level velocity controls and smooth force norms. Its signed nonlinear renewal must therefore fail a uniform absorption budget on the same forced construction. A Fourier multiplier estimate separates every low-frequency parent and every low renewal output. Low stretching outputs have a separate energy-paid history bound, leaving a target with every parent and both internal frequencies high. That target is unproved; no unforced regularity estimate is obtained.

1 An exact split and finite allowances

Let u, p, f solve

$$\partial_t u + (u\cdot\nabla)u = -\nabla p + \nu\Delta u + f, \qquad \nabla\cdot u = 0, \qquad \nu > 0,$$

on $\mathbb{R}^3 \times [a,T)$, where $0 < L = T - a < \infty$. Assume smoothness and, on each compact slab $[a,b] \subset [a,T)$, $u \in C^1([a,b]; H^m(\mathbb{R}^3))$ and $f, \nabla p \in C([a,b]; H^m(\mathbb{R}^3))$ for every integer $m \geq 0$. These hypotheses justify the following strong energy identities and integrations by parts. Write

$$E(t) = \|u(t)\|_2^2, \quad Y(t) = \|\nabla u(t)\|_2^2 = \|\omega(t)\|_2^2, \quad \omega = \operatorname{curl} u.$$

Suppose

$$E(t) \leq M, \quad \|f(t)\|_2 \leq F_0, \quad \|\Delta f(t)\|_2 \leq F_\Delta \qquad (a \leq t < T).$$

All constants in this note are independent of the shorter endpoint t. The kinetic-energy identity gives

$$\int_a^t Y(s)\,\mathrm{d}s \leq B, \qquad B = \frac{E(a)}{2\nu} + \frac{L\sqrt{M}F_0}{\nu}. \tag{1}$$

Indeed, $E'/2 = -\nu Y + \langle u, f\rangle$ and $E(t) \geq 0$.

Fix an angular Fourier cutoff $K \geq 0$, and set

$$u_L = \mathbf{1}_{\{|\xi|<K\}}(D)u, \quad u_H = u - u_L, \quad \omega_H = \operatorname{curl} u_H, \quad S_H = \tfrac{1}{2}(\nabla u_H + \nabla u_H^{\mathsf{T}}).$$

These are orthogonal L^2 projections commuting with derivatives. Define the physical tail quantities

$$Z_H = \tfrac{1}{2}\,\|\omega_H\|_2^2\,, \qquad D_H = \|\nabla\omega_H\|_2^2 = \|\Delta u_H\|_2^2\,, \qquad \Pi_H = \int_{\mathbb{R}^3} \omega_H \cdot S_H\omega_H \,\mathrm{d}x.$$

In particular, Π_H uses the high field in all three factors. With $\Pi = \int \omega \cdot S\omega$, introduce

$$P_L = \langle -\Delta u_L, u \times \omega\rangle, \quad P_H = \langle -\Delta u_H, u \times \omega\rangle, \quad X_H = P_H - \Pi_H, \quad F_H = \langle -\Delta u_H, f\rangle.$$

Then $P_L + P_H = \Pi$, and the exact tail balance is

$$Z_H' = \Pi_H + X_H - \nu D_H + F_H. \tag{2}$$

The filtered velocity is not assumed to solve an autonomous Navier–Stokes equation: X_H retains every lower-frequency input.

Proposition 1. *Let $c_K = (K^3/(6\pi^2))^{1/2}$. Under the preceding assumptions,*

$$\int_a^t |F_H(s)|\,\mathrm{d}s \le L\sqrt{M}F_\Delta, \tag{3}$$

$$\int_a^t |X_H(s)|\,\mathrm{d}s \le 3Kc_K\sqrt{M}B + K^2c_KM\sqrt{LB} =: C_L. \tag{4}$$

Neither bound uses a gradient supremum or tail dissipation of the velocity.

Proof. Integration by parts gives $F_H = -\langle u_H, \Delta f\rangle$, so $|F_H| \le \sqrt{M}F_\Delta$. This proves (3).

For the unitary Fourier convention, Cauchy–Schwarz on the ball gives

$$\|\nabla u_L\|_\infty \le Kc_K\sqrt{E}, \qquad \|\Delta u_L\|_\infty \le K^2c_K\sqrt{E}.$$

Here the gradient uses the Frobenius norm, and $c_K^2 = (2\pi)^{-3}|B_K|$. The vector and matrix bounds follow by Cauchy–Schwarz before taking a component sum.

For matrices define the trilinear form

$$b(A,B,C) = \sum_{i,j,k} A_{ji}B_{kj}C_{ki}, \qquad |b(A,B,C)| \le |A|_F|B|_F|C|_F.$$

The integrated stretching identity is $\Pi = -\int b(\nabla u, \nabla u, \nabla u)$, and the same identity holds for the smooth solenoidal field u_H. Put $A = \nabla u$, $H = \nabla u_H$ and $G = \nabla u_L = A - H$. The exact telescoping identity

$$b(A,A,A) - b(H,H,H) = b(G,A,A) + b(H,G,A) + b(H,H,G)$$

and $\|H\|_2 \le \|A\|_2$ imply

$$|\Pi - \Pi_H| \le 3\,\|\nabla u_L\|_\infty\, Y \le 3Kc_K\sqrt{M}\,Y. \tag{5}$$

Also

$$|P_L| \le \|\Delta u_L\|_\infty\,\|u\|_2\,\|\omega\|_2 \le K^2c_KM\sqrt{Y}.$$

Now $X_H = \Pi - \Pi_H - P_L$. Integrate these two bounds and use (1) and $\int_a^t \sqrt{Y} \le \sqrt{LB}$. □

Corollary 2. *If a finite constant A satisfies*

$$\int_a^t (\Pi_H - \nu D_H)\,\mathrm{d}s \le A \qquad (a \le t < T), \tag{6}$$

then the complete gradient energy obeys

$$Y(t) \le 2Z_H(a) + 2A + 2C_L + 2L\sqrt{M}F_\Delta + K^2 M.$$

In particular, a half-dissipation budget implies this conclusion, but the strictly smaller fraction is unnecessary for this fixed-cutoff gradient-energy implication.

Proof. Integrate (2), apply Proposition 1, and use $Y = 2Z_H + \|\nabla u_L\|_2^2 \le 2Z_H + K^2 M$. □

For an unforced solution, one can take $M = E(a)$ and $B = E(a)/(2\nu)$. Thus the low-frequency allowance is controlled by the datum, viscosity, cutoff and horizon. This does not prove (6): its uniform signed high-frequency history remains the open estimate. Allowing the full dissipation simply exposes its relation to the gradient-energy continuation criterion; it is not a new coercive estimate.

2 Application to the actual forced construction

The whole-space construction in [1] has smooth compactly supported forcing, bounded kinetic energy, zero initial velocity, and unbounded speed as $t \uparrow 1$. On slabs exhausting $[1/2, 1)$, all hypotheses above hold with fixed M, F_0, F_Δ.

The module `MettapediaNS.ForcedSignedBudget` in the Lean library applies directly to the construction's `CandidateProperties`. It proves that its gradient energy cannot have a uniform bound: such a bound would control the order-two energy via the forced viscous estimate, and the H^2 embedding would bound speed. It also proves the full-field signed-production obstruction directly, with force work bounded by $6\sqrt{E_0(u)}\sqrt{E_2(f)}$ in the doubled energy convention, where $E_j(v) = \|\nabla^j v\|_2^2$ sums all ordered derivative components. The constructor theorem is imported at the pinned revision `f9e8bc5b38b6e212696e8a30e3e91517af887bbd`.

Corollary 3. *For every such candidate, every fixed finite $K \ge 0$, and every $\beta \le 1$,*

$$\sup_{1/2 \le t < 1} \int_{1/2}^t (\Pi_H(s) - \beta\nu D_H(s))\,\mathrm{d}s = +\infty.$$

Meanwhile $\int_{1/2}^1 |X_H|$ and $\int_{1/2}^1 |F_H|$ are finite.

Proof. For $\beta = 1$, a finite upper bound would give a uniform gradient-energy bound by Corollary 2, contradicting the preceding forced-candidate theorem. For $\beta \le 1$, the accumulated quantity only increases because $D_H \ge 0$. The other assertions follow from Proposition 1. □

Thus the signed tail self-interaction is the term that defeats the budget; an infinite direct forcing-work allowance is not the cause. This conclusion applies to the complete constructed solution, without replacing its oscillatory corrections by a finite numerical model.

Forcing is still essential to that constructed history. With its zero datum and zero force, the kinetic-energy identity forces the smooth finite-energy solution to remain zero. In the construction the force is the smoothly extended momentum residual of the designed field (Section 10 of [1]). Removing it while keeping that field would change the equation. Restarting an unforced evolution from a later nonzero slice is a different problem, whose subsequent history is not settled by this argument. A proposed unforced budget must therefore use more than bounded forcing norms or bounded direct force work.

3 Forcing in a caloric stretching functional

The preceding obstruction can also test a functional that records stretching beyond the instantaneous enstrophy. Let

$$A = -\Delta, \qquad H_s = e^{-\nu s A}, \qquad R = \operatorname{curl} A^{-1} P, \qquad \mathcal{M}(v) = R\big((\operatorname{curl} v \cdot \nabla) v\big),$$

where P is the whole-space Leray projection. The multiplier R is used on a divergence: $\mathcal{M}(v) = R \operatorname{div}(v \otimes \operatorname{curl} v)$. Thus its low-frequency singularity causes no L^2 difficulty. Parseval's identity gives $\|R \operatorname{div} Q\|_2 \le \|Q\|_2$, where the tensor norm is Frobenius. For solenoidal v,

$$\|\mathcal{M}(v)\|_2 \le \|v\|_\infty \|\nabla v\|_2 , \qquad \langle Av, \mathcal{M}(v)\rangle = \Pi(v). \tag{7}$$

Define the nonnegative functional

$$\mathcal{K}(u) = \frac{1}{\nu} \int_0^\infty \|\mathcal{M}(H_s u)\|_2^2 \, \mathrm{d}s. \tag{8}$$

It is finite for the smooth L^2 fields considered here: the integrand is bounded near zero and is $O(s^{-5/2})$ at infinity by heat smoothing.

The effective force in the projected velocity equation is $g = Pf$, not necessarily f itself. Suppose in addition that

$$\|g(t)\|_\infty \le Q_0, \qquad \|\operatorname{curl} g(t)\|_\infty \le Q_1 \qquad (a \le t < T).$$

These constants are finite for the smooth compactly supported force in the construction: P is bounded on every H^m, Sobolev embedding controls Pf, and $\operatorname{curl} Pf = \operatorname{curl} f$. No L^∞ boundedness of P is assumed.

Proposition 4. *Put $c_h = (8\pi)^{-3/4}$. For a fixed smooth solenoidal slice u and direction g as above, with $E = \|u\|_2^2$ and $Y = \|\nabla u\|_2^2$,*

$$|D\mathcal{K}(u)[g]| \le \frac{c_h}{\nu^2} \left(\frac{32}{3} Q_0 E^{3/4} Y^{3/4} + 16 Q_1 E^{5/4} Y^{1/4} \right). \tag{9}$$

Consequently the standing evolution satisfies, uniformly for $t < T$,

$$\int_a^t |D\mathcal{K}(u)[g]| \, \mathrm{d}s \le C_{\mathcal{K}} := \frac{c_h}{\nu^2} \left(\frac{32}{3} Q_0 M^{3/4} L^{1/4} B^{3/4} + 16 Q_1 M^{5/4} L^{3/4} B^{1/4} \right). \tag{10}$$

This payment spends no order-two velocity dissipation.

Proof. Use the auxiliary heat parameter $r = \nu s$, and write $w_r = e^{-rA} u$, $h_r = e^{-rA} g$. Differentiating both parent slots gives

$$D\mathcal{M}(w)[h] = R \operatorname{div}(h \otimes \operatorname{curl} w + w \otimes \operatorname{curl} h).$$

The multiplier bound, heat contraction and (7) imply

$$|D\mathcal{K}(u)[g]| \le \frac{2}{\nu^2} \int_0^\infty \|w_r\|_\infty \|\nabla w_r\|_2 \left(Q_0 \|\nabla w_r\|_2 + Q_1 \sqrt{E} \right) \mathrm{d}r.$$

The Gaussian kernel has L^2 norm $c_h r^{-3/4}$, and its Fourier multiplier gives

$$\|w_r\|_\infty \le c_h r^{-3/4} \sqrt{E}, \qquad \|\nabla w_r\|_2 \le \min\{\sqrt{Y}, r^{-1/2}\sqrt{E}\}.$$

When $E, Y > 0$, split each integral at $r_* = E/Y$. Direct integration yields

$$\int_0^\infty \|w_r\|_\infty \|\nabla w_r\|_2^2 \, \mathrm{d}r \le \frac{16}{3} c_h E^{3/4} Y^{3/4},$$
$$\sqrt{E} \int_0^\infty \|w_r\|_\infty \|\nabla w_r\|_2 \, \mathrm{d}r \le 8 c_h E^{5/4} Y^{1/4}.$$

These bounds also justify differentiation under the heat integral for the smooth slices. If E or Y vanishes, $u = 0$ as an L^2 whole-space field and the result is immediate. This proves (9). Finally apply Hölder in physical time, using $E \le M$, $\int Y \le B$, and the interval length at most L. □

Let $N(u) = -P((u \cdot \nabla)u)$ be the actual unforced velocity source, and set

$$\mathcal{G} = \frac{Y + \mathcal{K}}{2}, \qquad \mathcal{T} = D\mathcal{K}(u)[N(u)], \qquad \mathcal{W} = \nu \|Au\|_2^2 + \frac{1}{\nu} \|\mathcal{M}(u) - \nu Au\|_2^2 .$$

In particular $\mathcal{G} \ge Y/2$ and $\mathcal{W} \ge 0$. Heat translation in (8) gives $D\mathcal{K}(u)[-\nu Au] = -\|\mathcal{M}(u)\|_2^2/\nu$. Combine this with the actual forced enstrophy identity and expand the square in $\mathcal{W}$ to obtain the exact balance

$$2\mathcal{G}' + \mathcal{W} = \mathcal{T} + D\mathcal{K}(u)[g] + 2\langle Au, f\rangle. \tag{11}$$

Both extra force terms have absolutely bounded physical-time integrals: their sum is bounded by $C_{\mathcal{K}} + 2L\sqrt{M} F_\Delta$.

Corollary 5. *On every forced candidate of the preceding section, for every $\beta \le 1$,*

$$\sup_{1/2 \le t < 1} \int_{1/2}^t (\mathcal{T} - \beta \mathcal{W}) \, \mathrm{d}s = +\infty.$$

Proof. Integrating (11) expresses $\int(\mathcal{T} - \mathcal{W})$ as $2\mathcal{G}(t) - 2\mathcal{G}(1/2)$ minus a uniformly bounded force integral. The candidate's gradient energy has no uniform upper bound, and $\mathcal{G} \ge Y/2$. This proves the case $\beta = 1$; decreasing β only increases the integral since $\mathcal{W} \ge 0$. □

Thus the smooth force does not hide an infinite payment in the derivative of this quartic functional. Its signed nonlinear renewal is itself unbounded relative to its positive cost on the forced history. A proof of the corresponding unforced budget must exploit the zero-force evolution; estimating these explicit force terms is insufficient. The estimate does not bound $\mathcal{T}$: substituting $N(u)$ for g would require velocity-dependent supremum norms that are not controlled by the energy inequality.

4 Low parents and high-frequency caloric renewal

The caloric return admits a further frequency separation. Fix $K \ge 0$, write $v = P_{\ge K} u$, $\ell = u - v$, and retain $E = \|u\|_2^2$, $Y = \|\nabla u\|_2^2$, $D = \|\Delta u\|_2^2$. Choose a universal homogeneous Sobolev constant S such that $\|w\|_6 \le S \|\nabla w\|_2$ for real scalar, vector and matrix fields on $\mathbb{R}^3$. The constant c_K is as in Proposition 1.

Proposition 6. *If $E \le M$, then for every $\varepsilon > 0$,*

$$|\mathcal{T}(u) - \mathcal{T}(v)| \le \varepsilon \nu D + C_2 Y^2 + C_{3/2} Y^{3/2}, \tag{12}$$

where all constants depend only on K, ν, M, ε and S:

$$a_K = \frac{2c_K S^3}{\nu^2}, \qquad b_K = \frac{6c_K S^3}{\nu^2}, \qquad C_2 = a_K\sqrt{M} + \frac{3b_K^{4/3}M}{4(4\varepsilon\nu)^{1/3}}, \qquad C_{3/2} = a_K K M.$$

In particular, the two last terms in (12) are at most $2(C_2 Y + C_{3/2}\sqrt{Y})\mathcal{G}$. Their coefficient is integrable in physical time under the energy bounds.

Proof. We first bound the complete quartic Fourier kernel of $\mathcal{K}$. For ordered inputs a, b let $\mathcal{M}_b(a,b) = R((\operatorname{curl} a \cdot \nabla)b)$. At frequencies p, q with output $k = p + q$, its coefficient satisfies

$$|\widehat{\mathcal{M}_b}(p,q)| \le |p|\,|\widehat{a}(p)|\,|\widehat{b}(q)|. \tag{13}$$

Indeed, $p \times \widehat{a}(p)$ is perpendicular to p, so its dot product with q equals its dot product with k. This cancels the $|k|^{-1}$ multiplier of R. The zero output is zero as well. Thus no lower bound on the output frequency is needed.

In an ordered quartic pairing of two such sources, heat integration gives the denominator $\nu^2(|p|^2 + |q|^2 + |r|^2 + |s|^2)$. Its kernel is bounded by

$$\frac{|p|\,|r|}{\nu^2(|p|^2 + |q|^2 + |r|^2 + |s|^2)} \le \frac{1}{2\nu^2}. \tag{14}$$

All parent pairs and all interference between their heat rates are included.

For a real field a, let $a^{\#}$ be the real scalar field with Fourier transform $|\widehat{a}|$. It has the same L^2 norm and homogeneous Sobolev norms as a. Its physical values need not be nonnegative. The absolute ordered quartic pairing is bounded by $1/(2\nu^2)$ times the nonnegative Fourier convolution corresponding to $\int a^{\#}b^{\#}c^{\#}d^{\#}$. Applying Hölder and Sobolev after this Fourier majorization gives

$$|D\mathcal{K}(u)[h]| \le \frac{2}{\nu^2}\left\|u^{\#}\right\|_6^3 \|h\|_2 \le \frac{2S^3}{\nu^2} Y^{3/2}\|h\|_2. \tag{15}$$

There are four differentiated slots. Subtracting the derivatives at u and v telescopes each of the three other slots, giving twelve terms. Since v is a Fourier restriction of u, coefficient majorization yields

$$\begin{aligned} |(D\mathcal{K}(u) - D\mathcal{K}(v))[h]| &\le \frac{6}{\nu^2}\left\|\ell^{\#}\right\|_\infty \left\|u^{\#}\right\|_4^2 \|h\|_2 \\ &\le \frac{6c_K S^{3/2}}{\nu^2} E^{3/4}Y^{3/4}\|h\|_2. \end{aligned} \tag{16}$$

The coefficient comparison takes place inside the Fourier convolution; it does not assert a pointwise inequality between $|v^{\#}|$ and $|u^{\#}|$. Here we used $\left\|\ell^{\#}\right\|_\infty \le c_K\sqrt{E}$ and interpolation $\left\|u^{\#}\right\|_4^2 \le S^{3/2}E^{1/4}Y^{3/4}$.

Write $\mathcal{B}(a,b) = -P((a\cdot\nabla)b)$. Then

$$N(u) - N(v) = \mathcal{B}(\ell, u) + \mathcal{B}(v, \ell), \qquad \|N(u) - N(v)\|_2 \le c_K\sqrt{EY} + Kc_K E.$$

Also Sobolev and interpolation give

$$\|N(v)\|_2 \le \|v\|_6\,\|\nabla v\|_3 \le S^{3/2}Y^{3/4}D^{1/4}.$$

Only the L^2 contraction and derivative commutation of the sharp projector are used; no general L^p bound for the ball multiplier is assumed. Split the return exactly as

$$\mathcal{T}(u) - \mathcal{T}(v) = D\mathcal{K}(u)[N(u) - N(v)] + (D\mathcal{K}(u) - D\mathcal{K}(v))[N(v)].$$

Equations (15)–(16) bound these terms by

$$a_K(E^{1/2}Y^2 + KEY^{3/2}) + b_K E^{3/4}Y^{3/2}D^{1/4}.$$

Young's inequality with exponents 4 and 4/3 bounds the last term by $\varepsilon\nu D + 3b_K^{4/3}EY^2/(4(4\varepsilon\nu)^{1/3})$. Use $E \le M$ and $Y \le 2\mathcal{G}$ to finish. □

Low renewal outputs are paid separately, even when both source parents are arbitrarily high. Set

$$g_K = P_{<K}N(v), \qquad \mathcal{T}_K(u) = D\mathcal{K}(v)[P_{\ge K}N(v)].$$

The full source $N(v)$ is formed before splitting its output. Cauchy–Schwarz in the unitary Fourier convolution gives $|\widehat{N(v)}(\xi)| \le (2\pi)^{-3/2}|\xi|E$. Integrating this over the ball, with an extra $|\xi|$ for curl, proves

$$\|g_K\|_\infty \le \frac{K^4 E}{8\pi^2}, \qquad \|\operatorname{curl} g_K\|_\infty \le \frac{K^5 E}{10\pi^2}. \tag{17}$$

The derivative is the output frequency because the nonlinearity is in divergence form; this estimate assumes no bound on parent gradients. Proposition 4, applied at the slice v, therefore gives

$$\int_a^t |D\mathcal{K}(v)[g_K]|\,\mathrm{d}s \le C_{\mathrm{out}}, \tag{18}$$

where C_{out} is the right side of (10) with $Q_0 = K^4M/(8\pi^2)$ and $Q_1 = K^5M/(10\pi^2)$. Although v is not an autonomous solution, its energy and integrated gradient energy are bounded by those of u, which is all that this application uses.

Corollary 7. *Suppose $A_0 \ge 0$ and $\theta < 1$ satisfy, on every shorter prefix,*

$$\int_a^t \mathcal{T}_K(u)\,\mathrm{d}s \le A_0 + \theta\int_a^t \mathcal{W}(u)\,\mathrm{d}s. \tag{19}$$

Choose $0 < \varepsilon < 1-\theta$, and let $C_f = C_{\mathcal{K}} + 2L\sqrt{M}F_\Delta$ for the forced equation, or $C_f = 0$ when $f = 0$. Then

$$\mathcal{G}(t) \le \left(\mathcal{G}(a) + \frac{A_0 + C_{\mathrm{out}} + C_f}{2}\right)\exp\Big(C_2B + C_{3/2}\sqrt{LB}\Big).$$

Proof. The exact split is $\mathcal{T}(u) = \mathcal{T}_K(u) + \mathcal{T}(u) - \mathcal{T}(v) + D\mathcal{K}(v)[g_K]$. Insert this in (11), integrate, and use Propositions 4 and 6. Since $\nu D \le \mathcal{W}$, the remaining cost has coefficient $1 - \theta - \varepsilon > 0$ and can be dropped. Grönwall applies with coefficient $C_2Y + C_{3/2}\sqrt{Y}$, whose integral is at most $C_2B + C_{3/2}\sqrt{LB}$. □

For an unforced regularity argument, K and A_0 in (19) must be chosen from the full initial datum, viscosity and a prescribed finite horizon, uniformly over every shorter lifespan and prefix. That high-frequency signed-history estimate is not proved here. The result pays low parents and low outputs; it does not pay repeated high-frequency regeneration.

The same estimates also sharpen the forced test: on every actual forced candidate, for each fixed K and every $\theta < 1$,

$$\sup_{1/2\le t<1} \int_{1/2}^{t} (\mathcal{T}_K(u) - \theta \mathcal{W}(u))\,\mathrm{d}s = +\infty.$$

A finite upper bound would give a uniform gradient-energy bound by Corollary 7, contradicting the constructed candidate. Unlike Corollary 5, this argument requires $\theta < 1$ because the low-parent estimate spends an arbitrarily small, positive part of the dissipation.

5 Low stretching outputs along the actual history

The output of $\mathcal{M}$ is distinct from the output of the nonlinear direction N. It too can be separated without restricting its parents. Fix $J \ge 0$, put $Q_- = P_{<J}$, $Q_+ = P_{\ge J}$, and define

$$\mathcal{K}_\pm(u) = \frac{1}{\nu}\int_0^\infty \|Q_\pm \mathcal{M}(H_s u)\|_2^2\,\mathrm{d}s, \qquad \mathcal{Z}_\pm(u) = \frac{1}{\nu}\|Q_\pm \mathcal{M}(u)\|_2^2, \qquad \mathcal{T}_\pm(u) = D\mathcal{K}_\pm(u)[N(u)].$$

Orthogonality gives $\mathcal{K} = \mathcal{K}_- + \mathcal{K}_+$ and $\mathcal{T} = \mathcal{T}_- + \mathcal{T}_+$. All source outputs are formed before the projection is applied.

Proposition 8. *With $c_J^2 = J^3/(6\pi^2)$,*

$$0 \le \mathcal{K}_-(u) \le \frac{c_J^2 E^2}{4\nu^2}, \qquad 0 \le \mathcal{Z}_-(u) \le \frac{c_J^2 EY}{\nu}. \tag{20}$$

Along the actual forced evolution,

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{K}_-(u) = -\mathcal{Z}_-(u) + \mathcal{T}_-(u) + D\mathcal{K}_-(u)[g]. \tag{21}$$

Consequently, uniformly for all shorter prefixes,

$$\left|\int_a^t \mathcal{T}_-(u)\,\mathrm{d}s\right| \le \frac{c_J^2 M^2}{4\nu^2} + \frac{c_J^2 MB}{\nu} + C_{\mathcal{K}}.$$

The last term is zero for an unforced evolution. This bounds the signed primitive, not the integral of $|\mathcal{T}_-|$.

Proof. The ordered bound (13) and Cauchy–Schwarz in the full convolution give, at every nonzero output k,

$$|\widehat{\mathcal{M}(u)}(k)| \le (2\pi)^{-3/2}\int |p|\,|\widehat{u}(p)|\,|\widehat{u}(k-p)|\,\mathrm{d}p \le (2\pi)^{-3/2}\sqrt{EY}.$$

Integrating its square over $|k| < J$ proves the $\mathcal{Z}_-$ bound. No upper bound on the parent frequencies was used. If $E_s = \|H_s u\|_2^2$ and $Y_s = \|\nabla H_s u\|_2^2$, then $E_s' = -2\nu Y_s$, so

$$\mathcal{K}_-(u) \le \frac{c_J^2}{\nu}\int_0^\infty E_s Y_s\,\mathrm{d}s \le \frac{c_J^2 E^2}{4\nu^2}.$$

Heat translation gives $D\mathcal{K}_-(u)[-\nu Au] = -\mathcal{Z}_-(u)$. The actual equation therefore proves (21). Proposition 4 applies to $\mathcal{K}_-$ with unchanged constants: inserting an orthogonal output projection in both factors of its derivative can only improve the two L^2 norm bounds. Integrate the identity, use $|\mathcal{K}_-(u(t)) - \mathcal{K}_-(u(a))| \le c_J^2 M^2/(4\nu^2)$, and use $\int \mathcal{Z}_- \le c_J^2 MB/\nu$. □

For the unforced equation there is a sharper payment, independent even of the horizon. The actual kinetic-energy equality is $E' = -2\nu Y$, so (20) gives

$$\mathcal{K}_-(u(t)) + \int_a^t \mathcal{Z}_-(u)\,\mathrm{d}s \le \frac{c_J^2 E(a)^2}{4\nu^2}. \tag{22}$$

Indeed, the bound for the integral is $c_J^2(E(a)^2 - E(t)^2)/(4\nu^2)$, which complements the bound for the remaining capacity. Therefore every unforced prefix satisfies

$$-\mathcal{K}_-(u(a)) \le \int_a^t \mathcal{T}_-(u)\,\mathrm{d}s \le \frac{c_J^2 E(a)^2}{4\nu^2} - \mathcal{K}_-(u(a)).$$

This is a bound on repeated low-output renewal from the initial kinetic energy; its constant grows with the fixed output cutoff.

There is a more direct way to use this payment in the regularity target. Keep the original, full positive cost $\mathcal{W}$, and set $\mathcal{G}_+ = (Y + \mathcal{K}_+)/2 \ge Y/2$. Subtracting (21) from (11) gives the exact identity

$$2\mathcal{G}_+' + \mathcal{W} = \mathcal{T}_+ + \mathcal{Z}_- + D\mathcal{K}_+(u)[g] + 2\langle Au, f\rangle. \tag{23}$$

Thus only the spent low-output cost, bounded by $C_{\mathrm{src}} = c_J^2 MB/\nu$, is needed on the right. The endpoint $\mathcal{K}_-$ allowance is unnecessary in this identity. In the unforced case one may take $C_{\mathrm{src}} = c_J^2 E(a)^2/(4\nu^2)$ by (22).

With the input cutoff K from the preceding section, define

$$v = P_{\ge K} u, \qquad \mathcal{T}_{K,J}(u) = D\mathcal{K}_+(v)[P_{\ge K} N(v)].$$

The estimates of Proposition 6 hold with $\mathcal{K}_+$ in place of $\mathcal{K}$: its ordered quartic kernel is restricted at the common output, and has the same bound. The force estimate and the low nonlinear-output payment hold as well. Consequently, if fixed $A_0 \ge 0$ and $\theta < 1$ obey

$$\int_a^t \mathcal{T}_{K,J}(u)\,\mathrm{d}s \le A_0 + \theta \int_a^t \mathcal{W}(u)\,\mathrm{d}s \tag{24}$$

on every shorter prefix, the same Grönwall argument gives

$$\mathcal{G}_+(t) \le \left(\mathcal{G}_+(a) + \frac{A_0 + C_{\mathrm{src}} + C_{\mathrm{out}} + C_f}{2}\right) \exp\left(C_2 B + C_{3/2}\sqrt{LB}\right).$$

Here $0 < \varepsilon < 1 - \theta$ defines $C_2, C_{3/2}$ as before. The exact decomposition used in (23) is

$$\begin{aligned}\mathcal{T}_+(u) = \mathcal{T}_{K,J}(u) &+ D\mathcal{K}_+(u)[N(u) - N(v)] \\ &+ (D\mathcal{K}_+(u) - D\mathcal{K}_+(v))[N(v)] + D\mathcal{K}_+(v)[P_{<K} N(v)].\end{aligned}$$

In particular, no autonomous evolution is assigned to the filtered field v.

Taking $J = K$ leaves a quintic term whose five original velocity inputs, nonlinear-direction output, and stretching output are all above the same fixed cutoff. Its physical-time bound (24) remains open, with the same initial-data and shorter-lifespan quantifiers as (19). On the actual forced construction it fails for every fixed K, J and every $\theta < 1$: a finite upper allowance would bound $\mathcal{G}_+ \ge Y/2$, contrary to the candidate's unbounded gradient energy. This further separation pays low stretching outputs; it does not bound regeneration among the remaining high frequencies.

6 Verification boundary

The Lean theorem `openAI_solution_no_uniform_signed_production_budget` applies to the actual construction at every positive viscosity and concerns the full-field spatial cubic integral. The sharp Fourier-tail extension and all caloric results here are analytic proofs, not Lean theorems or axioms. The companion `labs/caloric_force.py` checks the complete caloric balance by exact rational Fourier arithmetic on the periodic cube, including forces with a gradient part. It also checks the three-part frequency split, including a nonzero low-output contribution, and the ordered mode bound on nearly cancelling parents. Separate source-output checks retain the original positive cost and include arbitrarily high parent frequencies that feed low stretching outputs. These checks do not certify the whole-space estimates or a physical-time bound. Arbitrary-data unforced regularity remains open.

References

[1] OpenAI, *Finite time blowup for Navier–Stokes*, manuscript, with formalization in NavierStokesAndEuler, 2026.